

Flood Risk Assessment

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Architectural Drawing and engineering Service Submitted to Council to accompany an application.

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Flood Risk Assessment (FRA) Site: 17 Northgate Close, Balby, Doncaster, ZZ92 8CM Proposal: Change of use from 1 × C3 dwelling to 2 × self-contained C3 flats with a modest first-floor rear extension Scope and Planning Context This Flood Risk Assessment supports a Full Planning Application for redeveloping a residential property into a two-flat unit with an additional rear extension. The site is located on an established residential street, and the proposal does not introduce a more vulnerable land use than the existing lawful residential use. The development is therefore fully aligned with the principles of the National Planning Policy Framework (NPPF) and the Doncaster Strategic Flood Risk Assessment (SFRA). The proposal satisfies the Sequential Test because it involves the reuse of existing residential land within the urban area and does not require application of the Exception Test. The assessment demonstrates that flood risk has been appropriately considered, avoided where possible, and mitigated through proportionate design measures, in accordance with Doncaster Local Plan policy expectations. FRA focuses on: * Our methodology involves collecting and analyzing flood risk data from credible sources, including the Environment Agency, and integrating it into a high–medium–low probability and consequence matrix. This approach ensures clear planning by ranking flood risk sources, including fluvial, surface water, groundwater, and reservoirs. * Identification of the dominant flood risk pathway and its implications for design. * Mitigation and drainage principles, including future climate change allowances. <redacted_organisation>

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<redacted_date> <redacted_url> Site Description The site comprises an existing two-storey terraced dwelling within a built-up residential area of Doncaster. The surrounding context is entirely residential. The proposed works are predominantly internal, involving subdivision into two self-contained flats, with a small rear extension at first-floor level. The scheme retains the established residential character and does not materially increase the ground-level building footprint beyond typical domestic curtilage works. Flood Risk Sources – Summary (Environment Agency Long-Term Flood Risk) The Environment Agency's Long-Term Flood Risk information (downloaded on October 15, <redacted_date>) for 17 Northgate Close, Balby, Doncaster, ZZ92 8CM indicates the following: Surface Water (Pluvial / Flash Flooding) * Current risk: Low yearly probability * Future risk (<redacted_date>–<redacted_date>): Medium yearly probability (climate change scenario) Surface water flooding is identified as the primary flood risk pathway affecting the site. Under extreme rainfall events, shallow surface water accumulation is possible. For assessment purposes, envision this as 'ankle-deep water,' a peak runoff depth in the order of 150–200 mm, considered a reasonable worst-case design assumption during intense storm events. Rivers and the Sea (Fluvial / Tidal) * Current risk: Low yearly probability * Future risk (<redacted_date>–<redacted_date>): Low yearly probability Groundwater * The site lies outside a groundwater flood alert area. * Probability and consequence are considered low. Reservoirs * Flooding from reservoirs is considered a very low probability and unlikely. <redacted_organisation>

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<redacted_date> <redacted_url> Conclusion: The dominant risk to the site is surface water flooding, particularly under future climate change scenarios. Fluvial flood risk is considered low, with an annual probability of less than 1% (classified in the low probability band) now and under future projections. For clarity, a “medium yearly chance” of surface water flooding may be interpreted as approximately a 5–10% annual probability during the <redacted_date>–<redacted_date> period, underscoring the importance of appropriate surface water management rather than site avoidance. Vulnerability and Flood Zone Considerations The existing and proposed uses are residential (Use Class C3); therefore, the development does not increase flood vulnerability. No basement or below-ground accommodation is proposed. The scale of development is modest and, with appropriate drainage design, will not increase flood risk elsewhere. While formal Flood Zone confirmation should be verified via the Environment Agency Flood Map for Planning, the available long-term risk data indicates low fluvial risk, consistent with a low-risk fluvial context. Impact of the Development on Flood Risk Elsewhere The development is designed to ensure that post-development runoff will be no worse than existing, and where feasible, improved. To verify and sustain the targeted 30% reduction in surface water runoff rates, a monitoring plan will be implemented. This plan will include bi-annual inspections of the drainage system and data collection on runoff rates, which will be compared against baseline pre-construction data to

ensure compliance. Adjustments to the system will be made as necessary to achieve and uphold the targeted reduction. As a design objective, the scheme will aim to: Achieve a minimum 30% reduction in surface water runoff rates, or Match greenfield equivalent runoff rates, where practical. This measurable target converts policy intent into a clear, deliverable commitment, ensuring the development enhances rather than burdens local drainage infrastructure. Key principles: Retain existing drainage routes. Control peak discharge during heavy rainfall events. Ensure the rear extension does not generate uncontrolled runoff from additional hardstanding. Finished Floor Levels and Flood Resilience Given the identified future surface water risk, the following resilience measures are proposed: Finished floor levels: Existing internal floor levels retained; no lowering proposed. Thresholds: External door thresholds maintained or raised where feasible, typically 150 mm above external ground level, without compromising accessibility. <redacted_organisation>

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